STD IX
PHYSICS
WS -2

THIRUVANANTHAPURAM EDUCATIONAL DISTRICT
RAINBOW PHYSICS TEACHERS GROUP - STD IX

EQUATIONS OF MOTION

- Understanding Motion with Key Concepts:
  - Distance and Displacement:

    Distance refers to the total length of the path covered, while displacement is the straight-line distance between an object's initial and final positions in a definite direction.

  - Velocity:

    Velocity is the displacement of an object in unit time. It is a vector quantity, meaning it has both magnitude (speed) and direction. The unit of velocity is m/s.

  - Uniform and Non-uniform Velocity:

    An object has uniform velocity if its magnitude of displacement is equal at equal intervals of time while moving in the same direction. Conversely, non-uniform velocity occurs if either the speed or the direction (or both) of an object's motion changes.

  - Acceleration and Retardation:

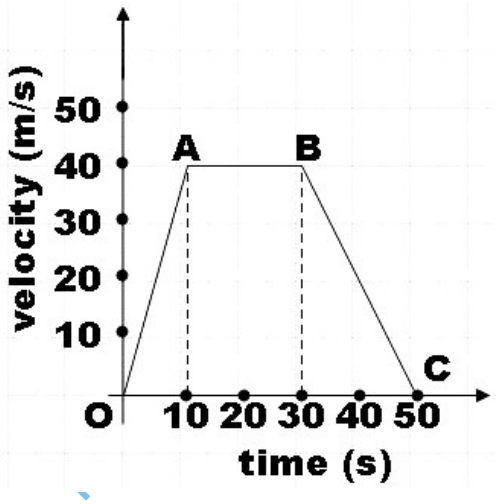
    Acceleration is the change in velocity of an object in unit time, or the rate of change of velocity. It is also a vector quantity. When velocity decreases, it is termed negative acceleration or retardation. Its unit is also m/s^2.

- Application in Road Safety and Accident Prevention:
  - Controlling Velocity and Acceleration.
  - Strictly following traffic rules is essential to reduce accidents.
  - Pedestrians also play a crucial role in road safety by following motion-based rules. They should walk along the right side of the road and cross only at designated zebra crossings, obeying traffic signals. Wearing light-coloured clothes in dim light is advisable as it increases visibility for drivers, reducing the risk of accidents caused by reduced perception of pedestrians.

RAINBOW PHYSICS TEACHERS GROUP

- Road signs and road markings are implemented to reduce road accidents and ensure safety. They provide crucial information about expected motion, speed changes, and potential hazards:
    - Mandatory Signs enforce compulsory rules, such as "No Parking" or "No Bicycles", guiding permissible movements.
    - Cautionary Signs warn drivers about upcoming road conditions like "Hump" or "Narrow Bridge," prompting them to adjust their velocity or acceleration.
    - Informatory Signs provide details like directions or distances to facilities, which aid in planning routes and maintaining a steady flow of traffic.
    - Road Markings, such as "Zebra crossings," define safe areas for specific movements, like pedestrian crossings or where vehicles can cross.

1.Graph related to the motion of a car is given below.



a) What type of motion is represented by OA ?
b) What type of motion is represented by AB ?
c) What type of motion is represented by BC ?
d) Find the acceleration of the car from O to A ?
e) Find out the distance travelled by the car from A to B ?

2.Classify the situations given below as uniform and non-uniform velocity

a) The velocity of a car runs along a straight road in the same direction with the same speed

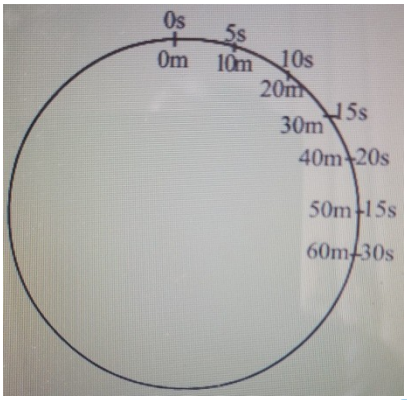
TEACHERS GROUP

b) A car runs along a curved road
c) Motion of a stone dropped from a height
d) When light travels through vacuum
e) Swinging on a swing
f) A ball rolling on a ground

3.The distance travelled by three cars A,B and C are

a)Car A	0m	10m	20m	30m	40m	50m	60m
	0s	5s	10s	15s	20s	25s	30s
b)Car B	0m	10m	20m	30m	40m	55m	60m
	0s	5s	10s	15s	20s	25s	30s

c)Car C



Vehicle	Uniform/Non Uniform Velocity	Reason
Car A		
Car B		
Car C		

4.For each question, choose the correct option:

- (A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.
- (B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.
- (C) Assertion is true, but reason is false.
- (D) Assertion is false, but reason is true.

RAINBOW PHYSICS TEACHERS GROUP

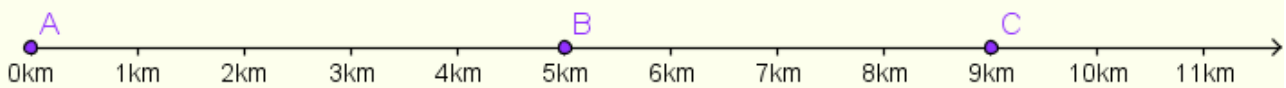
(i)Assertion**:** A car moves in a circular track and comes back to the starting point. The displacement is zero.

Reason**:** Displacement is a vector quantity.

(ii)Assertion: The magnitude of displacement is always equal to the distance travelled.

Reason: Displacement is a scalar quantity.

5.An object moves from point A to point B to point C, then back to point B and then to point C along the line shown in the figure below.



a)If the object had moved directly from point A to point C without turning back, what would have been the displacement and distance?

b)What is the total distance travelled by the object?

c) What is the displacement between point A and point B?

d)Which of the following is true for this motion?

- A) Distance = Displacement
- B) Distance > Displacement
- C) Distance < Displacement
- D) Displacement is negative

6.Which among the following is a characteristic of uniform acceleration ?

a) Constant velocity

b) Constant rate of change of velocity

c) Zero acceleration

d) Changing acceleration

7.What is the purpose of a zebra crossing?

a) To indicate a parking zone

b) To provide safe space for pedestrians to cross the road

c) To designate a bus stop

d) To allow vehicles to overtake

8.Rate of change of velocity is acceleration

a) What is the SI unit of acceleration?

RAINBOW PHYSICS TEACHERS GROUP

b)Differentiate between uniform acceleration and non uniform acceleration.

9.A bus accelerates uniformly from 54 km/hr to 72 km/hr in 10 seconds.

a) Calculate the acceleration in m/s^2.

b) Distance covered by the bus in meters during the interval.

10.Match the following

	
	Cautionary Sign
	
Informatory Sign	
	
Mandatory sign	

11 a) Why road signs and road markings are implemented ?

b) Draw the sign corresponding to no parking

12.What does the following sign indicate? What category of road sign does it belong?



RAINBOW PHYSICS TEACHERS GROUP

13.a) Identify the road markings given below



b)Give any two traffic rules to be followed by pedestrians.

14.Say whether the statements are true or false
a)Speed and velocity are same physical quantity
b)A car moving in a circular path with constant speed has uniform velocity
c)Velocity can be negative.
d)A vehicle can have different velocities even if its speed is the same

15. For each question, two statements are given: an Assertion (A) and a Reason (R). Choose the correct option from the choices given below:
a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, but R is not the correct explanation of A.
c) A is true, but R is false.
d) A is false, but R is true.
e) Both A and R are false

(i) Assertion (A): A vehicle moving along a circular path with a constant speed has non-uniform velocity.
Reason (R): If the direction of motion of an object changes, its velocity will also change, even if its speed remains constant.
(ii) Assertion (A): When a driver applies brakes to bring a vehicle to rest, the vehicle undergoes negative acceleration.
Reason (R): Retardation is defined as the rate of decrease in velocity, and its unit is m/s^2

RAINBOW PHYSICS TEACHERS GROUP